

RAKSHITA N

 rakshita1967@gmail.com  7708624230  Coimbatore  linkedin.com/in/rakshita-n  github.com/rakshita1802

Professional Summary

Software Systems student skilled in C++, and Python with a strong foundation in Data Structures, Algorithms, and AI. Experienced in developing software projects and applying problem-solving skills to real-world challenges. Quick learner with a passion for AI, software development, and emerging technologies.

Education

M. Sc. Software Systems (5 years integrated), Coimbatore Institute of Technology CGPA: 8.3 [Till 4th sem]	2024 – 2029
Class X, Sri Chaitanya Techno School	2022 – 93%
Class XII, Sri Chaitanya Techno School	2024 – 81%

Projects

PetCare Management System

- Developed a full-stack web application using **FastAPI, Next.js, SQLite, and SQLAlchemy**, implementing RESTful APIs with CRUD operations, search, filtering, pagination, and sorting.
- Designed a relational database with foreign key constraints and implemented frontend validation, error handling, and state management using React Hooks.

University Knowledge Base | Python, LangChain, RAG, ChromaDB, Streamlit

- Developed an AI-powered University Knowledge Base using **Python, LangChain, RAG, ChromaDB, and Streamlit**, enabling context-aware question answering from university documents.
- Implemented document ingestion, text chunking, local embeddings, vector search, and OCR support for efficient knowledge retrieval.

Secure Digital Vault System | FastAPI, React, SQLite, 8086 Assembly

- Developed a secure digital vault for storing and managing sensitive data using authentication and encryption.
- Integrated a cryptographic module using 8086 Assembly with a FastAPI backend and React frontend for secure data management.

AgriHive AI | NIT Trichy Hackathon

- Built a privacy-preserving agricultural intelligence platform enabling collaborative learning across farms using **Feature-Aware Personalized Federated Learning (FAP-FL)**.
- Designed a modular AI pipeline integrating **Random Forest, Federated Learning, Digital Twin Simulation, PSO Optimization, and Explainable AI (SHAP)** for personalized agricultural risk assessment.
- Developed a scalable full-stack solution with **Python, FastAPI, React, PostgreSQL, SQLAlchemy**, and external weather APIs.

Technical Skills

Programming Languages

C, C++, Python, Java

Web Technologies

HTML, CSS, JavaScript, Bootstrap

Frameworks & Tools

FastAPI, Next.js, React, Git, GitHub

Artificial Intelligence

RAG, LangChain, Generative AI, Prompt Engineering

Databases

MySQL, SQLite, Vector Databases (Chroma)

Certificates

- Revature FullStack Development Certificate
- Revature Generative AI training program

Achievements

1st place Tata Centre for Artificial Intelligence and Machine Learning NIT Trichy

2nd Prize – Revature Capstone Project

Certificate of Excellence – Best Theme, Rotaract Club of Coimbatore Uptown
Rac-a-Thon Hackathon